Tuan N. Le

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Astronomy Ph.D. Applicant

EDUCATION

University of Science, VNUHCM (HCMUS) 2025 – present
M.Sc. in Theoretical Physics *GPA: 8.5/10*

- Thesis topic: Black hole mass measurement in the radio galaxy NGC 3862 (3C 264) using ALMA molecular-gas kinematics
- Advisor: Dr. Dieu D. Nguyen

University of Technology, VNUHCM (HCMUT) 2019 – 2023
B.Eng. in Mechanical Engineering *Thesis: 4.0/4.0*

RESEARCH INTERESTS

- Dynamical black hole mass measurements from stellar and gas kinematics in galaxy nuclei
- Black hole seeding and the low-mass end of the black hole population
- Active galactic nuclei and their effect on host galaxies
- Black hole growth, accretion, and coevolution with host galaxies across cosmic time

TECHNICAL SKILLS

Programming	Python, MATLAB, LaTeX
Python libraries	NumPy, Matplotlib, Astropy, emcee, AdaMet
Modeling software	Multi-Gaussian Expansion fitting (MGEfit) Penalized Pixel-Fitting (pPXF) Jeans Anisotropic Modelling (JAM) KINematic Molecular Simulation (KinMS) ELT/HARMONI SIMulator (HSIM)
Statistics	MCMC sampling, Bayesian inference, bootstrap model ensembles
Space-based data	JWST/NIRSpec IFU, HST imaging
Ground-based data	Keck/OSIRIS LGS-AO, ALMA
Simulated data	ELT/HARMONI integral-field spectroscopy

RESEARCH EXPERIENCE

Stellar-dynamical black hole masses: from SMBHs to IMBHs 2025 – present
Advisor: Dr. Dieu D. Nguyen *University of Michigan (remote)*

Measured dynamical black hole masses across the SMBH-to-IMBH regimes using resolved stellar kinematics to constrain black hole seeding mechanisms and low-mass end scaling relations.

- **M81 (NGC 3031), JWST/NIRSpec IFU** *Main contributor, ApJ 2026*
  - Separated the non-thermal AGN continuum from the stellar light, enabling an accurate stellar mass model and unbiased stellar kinematics
  - Extracted high-resolution 2D stellar kinematics from the JWST/NIRSpec IFU data
  - Modeled the kinematics with an ensemble of 24 Jeans models spanning M/L gradients, PSF, AGN masking, and orbital anisotropy, yielding $M_{\rm BH} = (4.77 \pm 0.37) \times 10^7\, M_\odot$ including systematics and resolving long-standing discrepancies among previous measurements
- **IC 342, Keck/OSIRIS LGS-AO** *Project lead, submitted to ApJ*
  - Probed a nucleus hidden behind heavy Galactic extinction, using K-band AO spectroscopy at $0.062''$ to resolve the small sphere of influence of an IMBH
  - Extracted 2D stellar kinematics from the CO bandheads, revealing a subtle central rise in velocity dispersion ($\sim$42 to 49 km s^{-1}) within the nuclear star cluster
  - Demonstrated that applying a mass-to-light ratio gradient from the young nuclear star cluster is critical for uncovering the central black hole signature, deriving a dynamical mass of $M_{\rm BH} \approx 2 \times 10^5\, M_\odot$.

Gas-dynamical black hole masses in AGN hosts galaxies 2025 – present

Advisor: Dr. Dieu D. Nguyen *University of Michigan (remote)*

Measured dynamical black hole masses in distant giant elliptical galaxies using ALMA molecular-gas kinematics, mitigating stellar modeling biases from bright nuclear and jet contamination.

- **NGC 3862 (3C 264), ALMA** *Project lead, submitted to ApJ*
  - Modeled the ^{12}CO(2–1) kinematics of a regularly rotating circumnuclear gas ring with KinMS, spatially resolving the sphere of influence
  - Built a clean stellar mass model from HST imaging by masking the synchrotron jet and AGN and correcting the circumnuclear dust extinction with an F475W–F814W color map
  - Measured $M_{\rm BH} \approx 2 \times 10^9\, M_\odot$ and a dynamical M/L aligns with an old, metal-rich stellar population

Black hole mass measurements with next-generation facilities 2023 – 2025

Advisor: Dr. Dieu D. Nguyen *Vietnam*

Feasibility studies for upcoming instruments, aimed at pushing dynamical black hole masses to lower masses regime and to larger distances.

- Ran mock ELT/HARMONI observations and MCMC dynamical fits to test the recovery of IMBH masses in nuclear star clusters (AJ, 2025)
- Explored the detectability of IMBHs in extragalactic globular clusters with ELT/HARMONI (presented at the 2023 SAGI Workshop)

RESEARCH POSITIONS

Research Assistant, Simons Astro-Physics Group at ICISE (SAGI) 06/2025 - 12/2025

Advisor: Dr. Dieu D. Nguyen *ICISE, Quy Nhon, Vietnam*

- Reduced JWST/NIRSpec IFU data of M81 and separated the non-thermal AGN continuum from the stellar light, enabling an accurate stellar mass model and the dynamical SMBH mass measurement
- Analyzed the stellar kinematics of the heavily dust-obscured spiral galaxy IC 342
- Performed feasibility simulations for JWST observing proposals (NIRSpec IFU and NIRCam)

SELECTED PUBLICATIONS

Submitted

1. **T. N. Le**, D. D. Nguyen, M. Durré, A. W. Graham, E. Gallo, H. N. Ngo, T. H. T. Ho, T. Q. T. Le, T. Hoang, I. V. Chilingarian, R. Soria, F. Zou, H. G. Tong, N. K. N. Le, Q. T. Le, H. M. B. Tran, *A $10^5\, M_\odot$ Black Hole in the Bulgeless Spiral Galaxy IC 342: First Stellar-Dynamical $M_{\rm BH}$ Measurement with Keck/OSIRIS LGS-AO*, submitted to ApJ (2026).

2. **T. N. Le**, D. D. Nguyen, H. N. Ngo, E. Gallo, T. Q. T. Le, T. H. T. Ho, F. Zou, Q. T. Le, H. G. Tong, N. K. N. Le, *Unveiling the Central Black Hole of NGC 3862: A Direct Mass Determination Using ALMA's High-Resolution Molecular Gas Kinematics*, submitted to ApJ (2026).

Peer-Reviewed

1. D. D. Nguyen, **T. N. Le** (main contributor), M. Cappellari, H. N. Ngo, T. Q. T. Le, T. H. T. Ho, L. Q. T. Nguyen, E. Gallo, F. Zou, M. Perna, N. Thatte, M. Pereira-Santaella, *The Supermassive Black Hole in the Nearby Spiral Galaxy M81: A Robust Mass from JWST/NIRSpec Stellar Dynamics*, ApJ 1003, 98 (2026). doi:10.3847/1538-4357/ae64f0

2. D. D. Nguyen, H. N. Ngo, M. Cappellari, T. Q. T. Le, T. H. T. Ho, **T. N. Le**, E. Gallo, N. Thatte, F. Zou, M. Perna, M. Pereira-Santaella, *Measuring the Central Dark Mass in NGC 4258 with JWST/NIRSpec Stellar Kinematics*, ApJ 999, 97 (2026). doi:10.3847/1538-4357/ae32f2

3. D. D. Nguyen, B. D. Boizelle, H. N. Ngo, E. Gallo, **T. N. Le**, S. Thater, T. H. T. Ho, T. Q. T. Le, Q. T. Le, S. Norcross, X. Li, H. G. Tong, N. K. N. Le, H. M. B. Tran, *ALMA ^{12}CO(2–1) Gas Dynamics in NGC 315: A Multimethod Benchmark for Supermassive Black Hole Mass Measurement*, ApJ 1010, 66 (2026). doi:10.3847/1538-4357/aea151

4. D. D. Nguyen, L. Q. T. Nguyen, E. Gallo, H. N. Ngo, Q. T. Le, F. Pacucci, T. Q. T. Le, **T. N. Le**, T. H. T. Ho, S. Thater, H. G. Tong, N. K. N. Le, *Dynamical Evidence for a Billion Solar-mass Black Hole in Galaxy NGC 4061 from ALMA ^{12}CO(2–1) Kinematics*, ApJ 1005, 94 (2026). doi:10.3847/1538-4357/ae771c

5. T. Q. T. Le, D. D. Nguyen, H. N. Ngo, T. H. T. Ho, **T. N. Le**, L. Q. T. Nguyen, *Probing the Variation of the Inner Surface-Brightness Profile of Nuclear Star Clusters on the Intermediate-Mass Black Hole Mass Measurements Using Mock Observations of ELT/MICADO and HARMONI*, Universe 12, 160 (2026). doi:10.3390/universe12060160

6. D. D. Nguyen, M. Cappellari, T. Q. T. Le, H. N. Ngo, E. Gallo, N. Thatte, F. Zou, T. H. T. Ho, **T. N. Le**, H. G. Tong, M. Pereira-Santaella, *Extending the Frontier of Spatially Resolved Supermassive Black Hole Mass Measurements to $1 \lesssim z \lesssim 2$: Simulations with ELT/MICADO High-Resolution Mass Models and HARMONI Integral-Field Stellar Kinematics*, MNRAS 546, 4 (2026). doi:10.1093/mnras/stag238

7. D. D. Nguyen, M. Cappellari, H. N. Ngo, T. Q. T. Le, **T. N. Le**, K. N. H. Ho, A. K. Nguyen, P. T. On, H. G. Tong, N. Thatte, M. Pereira-Santaella, *Simulating Intermediate Black Hole Mass Measurements for a Sample of Galaxies with Nuclear Star Clusters Using ELT/HARMONI High Spatial Resolution Integral-Field Stellar Kinematics*, AJ 170, 124 (2025). doi:10.3847/1538-3881/ade9ba

PRESENTATIONS

- **Seminar talk** – *Measuring Supermassive Black Hole Masses: Dynamical Studies from the Galaxy Dynamics Lab*. "Topics in Cosmology" Seminar, Faculty of Physics and Engineering Physics, University of Science, VNUHCM, Vietnam (Sep 2026).
- **Contributed talk** – *Uncovering Intermediate-Mass Black Holes in Extragalactic Globular Clusters using ELT/HARMONI High Spatial Resolution Integral-Field Stellar Kinematics*. 2023 SAGI Astrophysics Workshop on Dust Polarimetry and Applications in Astrophysics, ICISE, Quy Nhon, Vietnam (Nov 2023).
- **Contributed talk** – *Hunting Intermediate-Mass Black Holes with the Extremely Large Telescope*. New Frontiers of Astrophysics in the Golden Era of Dust Polarimetry (SAGI Workshop), ICISE, Quy Nhon, Vietnam (Jul 2022).

SCHOOLS & WORKSHOPS

48th International School for Young Astronomers (ISYA) 27 Sep – 16 Oct 2026
IAU / ICISE, Quy Nhon, Vietnam

- Three-week IAU school covering galaxies, AGN, cosmology, stellar evolution, interstellar medium, exoplanets, radio astronomy and machine learning in astronomy
- Hands-on training with Virtual Observatory tools (TOPCAT, Aladin, VizieR, PyVO) and machine-learning workshops on astronomical time-series and image data
- Team leader of the group project on flare activity of M dwarfs using Quy Nhon Observatory and TESS data

10th Vietnam School on Astrophysics (VSOA10): Cosmology, ICISE, Quy Nhon, Vietnam Jul 2026

9th Vietnam School on Astrophysics (VSOA9): Galaxies, ICISE, Quy Nhon, Vietnam Jul 2025

LEADERSHIP & OUTREACH

University of Science Astronomy Club (USAC) 2019 – present
Student astronomy club at University of Science, VNUHCM *Ho Chi Minh City, Vietnam*

- **Research Mentor** (2022 – present)
  Guiding student teams through their annual astronomy research presentations.
  Offering academic feedback on scientific content and presentation skills.
- **Head of Content Team** (2020 – 2022): Organizing astronomy talks within the club and bringing them to the wider public through workshops and seminars.

Host & Moderator, *The PhD Application Journey in Astronomy* Aug 2026
Online talkshow series for Vietnamese students applying to astronomy graduate programs

REFERENCES

Dr. Dieu D. Nguyen
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